Components, Ohm's law & resistor networks

National 5 · Electronics & control · Answers

Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 Calculate the total resistance.

- 1/RT = 1/470 + 1/470
- RT = 235 Ω

3 Quick check

- a) 500 Ω
- b) 3 V
- c) 0.018 A
- d) 230 V a.c.

4 Key terms

- 1 – B
- 2 – F
- 3 – E
- 4 – A
- 5 – D
- 6 – C

5 Calculate the power of the lamp.

- P = IV = 0.5 × 12
- P = 6 W

6 Draw the symbol for an ammeter and state how it is connected.

- Circle with A
- Connected in series with the lamp

7 State the effect on V1 and on the current.

- V1 increases
- The current decreases